PRINCIPLES OF PLASMA PROCESSING
Course Notes: Prof. F.F. Chen

PART A7: PLASMA DIAGNOSTICS

X. INTRODUCTION

Diagnostics and *sensors* are both measurement methods, but they have different connotations. Diagnostic equipment is used in the laboratory on research devices and therefore can be a large, expensive, and one-of-a-kind type of instrument. Sensors, on the other hand, are used in production and therefore have to be simple, small, unobtrusive, and foolproof. For instance, *end-point detectors*, which signal the end of an etching step by detecting a spectral line characteristic of the underlying layer, are so important that they are continually being improved. Practical sensors are few in number but constitute a large subject which we cannot cover here. We limit the discussion to laboratory equipment used to measure plasma properties in processing tools.

Diagnostics for determining such quantities as n, KT_e, V_s, etc. that we have taken for granted so far can be remote or local. Remote methods do not require insertion of an object into the plasma, but they do require at least one window for access. Local diagnostics measure the plasma properties at one point in the plasma by insertion of a probe of one type or another there. Remote methods depend on some sort of radiation, so the window has to be made of a material that is transparent to the wavelength being used. Sometimes quartz or sapphire windows are needed. The plasma can put a coating on the window after a while and change the transmission through it. Probes, on the other hand, have to withstand bombardment by the plasma particles and the resulting coating or heating; yet, they have to be small enough so as not to change the properties being measured.

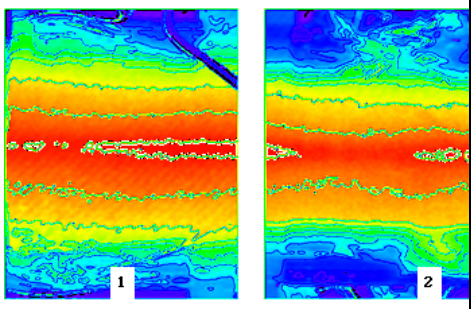


Fig. 1. Emission of ionized argon light at various z positions in a helicon discharge.

XI. REMOTE DIAGNOSTICS

1. Optical spectroscopy

One common remote diagnostic is optical emission spectroscopy (OES), which is the optical part of the more general treatment of radiation covered in Part B. In OES, visible light is usually collected by a lens and focused onto the slit of a spectrometer. The detector can be a photodiode, a photomultiplier, or an optical multichannel analyzer (OMA). With a photodiode, interference filters are used to isolate a particular spectral line. Optical radiation can also be used to image a plasma in the light of a particular spectral line using an interference

filter and a sensitive CCD camera. Fig. 1 shows the emission from ionized argon recorded with a narrow-band filter for the 488 nm line of Ar^+.

A photomultiplier can see only one part of the spectrum at a time, but it is the most sensitive detector for faint signals. An OMA records an entire range of wavelengths on a CCD (charge-coupled detector) and is the convenient for scans of a single line or for recording an entire spectrum.

By comparing the intensities of different spectral lines, one can determine not only the atomic species present but also the electron temperature, density, and the ionization fraction. The relative intensities of two lines with different excitation thresholds can yield KT_e. The relative intensities of an ion line and a neutral line can be used to estimate the ionization fraction. In principle, line broadening contains a large amount of information, but only for hot, highly ionized plasmas. For instance, *Doppler broadening* yields the velocity of the emitting ion or atom. *Stark broadening* or *pressure broadening* gives information on density. This is because, at high densities, collisions interrupt the emission of radiation, and hence the line cannot contain a single frequency. In plasma processing, the most useful and well developed technique is *actinometry*. In this method, a known concentration of an impurity is introduced, and the intensities of two neighboring spectral lines, one from the known gas and one from the sample, are compared. Since both species are bombarded by the same electron distribution and the concentration of the actinometer is know, the density of the sample can be calculated.

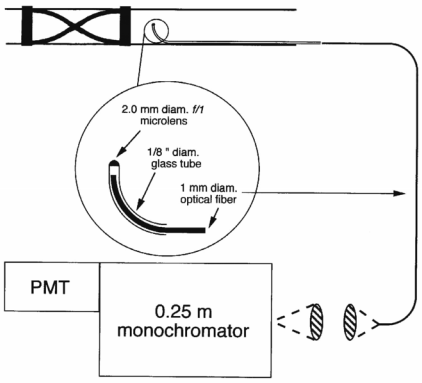


Fig. 2. Schematic of a local OES probe.

Though most optical methods average over a ray path in the plasma, a more local measurement of light emission can be made with a probe containing a small lens coupled to an optical fiber. Such a probe is shown in Fig. 2, and data from it in Fig. 3.. The lens collects light preferentially from a small focal spot just in front of it. The Ar^+ light collected by it is localized under the antenna if $\mathbf{B}_0 = 0$, as would be expected in ICP operation.

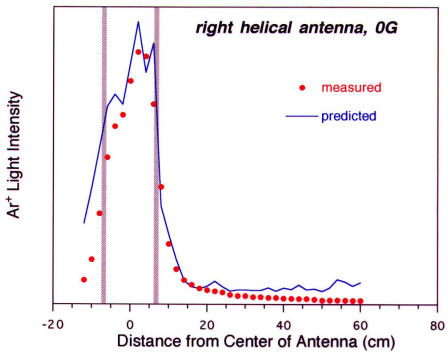


Fig. 3 Example of data on optical emission vs. *z*.

2. Microwave interferometry

Another useful remote diagnostic is *microwave interferometry*. A beam of microwave radiation is launched by a horn antenna into a plasma through a window. According to Eq. (A5-1), these waves can propagate in the plasma if $\omega > \omega_p$. From (A5-1) it is easily seen that the phase velocity in the plasma is

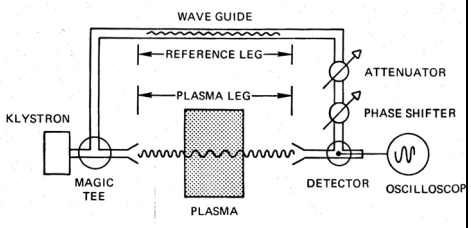


Fig. 4. Schematic of a microwave interferometer (Chen, p. 91).

$$\frac{\omega}{k} = \frac{c}{(1-\omega_p^2/\omega^2)^{1/2}}. \qquad (1)$$

This is faster than the velocity of light, but it is quite all right for *phase* velocity to be > *c* as long as the group velocity is < *c*. The microwave beam therefore has a longer wavelength inside the plasma than in air. The presence of the plasma therefore changes the phase of the microwave signal, a change which increases with the density of the plasma. The "standard" setup is shown in Fig. 4. The microwave beam from a generator is split into two parts, one going through the plasma and the other going through a waveguide toward the detector, where the two beams are recombined By adjusting the reference signal with an attenuator and phase shifter, the two signals can be made to cancel each other, so that the detector shows zero signal when there is no plasma. If the plasma density is increased slowly, the signal going through the plasma will have undergone fewer oscillations, and this phase shift will cause the nulled detector to give a finite dc signal output. If the plasma density reaches a value such that the wave loses exactly one wavelength, the detector will again return to zero; the signal is shifted by one *fringe*. By counting the number of fringes either on the way up to maximum density or on the way down, one obtains a measure of the average density traversed by the microwave beam. Though this illustrates the principle of interferometry, it is not normally done this way. First, ω is usually chosen so that ω_p/ω is a small number; then, the phase shift is linearly proportional to *n*. Second, the entire reference leg can be replaced by a mirror on the opposite side of the plasma to reflect the beam back into the launching horn. The beam then travels twice through the plasma and suffers twice the phase shift. Besides increasing the sensitivity, this method obviates phase shifts in the reference leg due to small changes in room temperature, which change the length of the waveguide. If the plasma is always on, it is difficult to set the initial null of the detector. There are various ways to get around this which we need not explain here. Modern network analyzers can do most of these calibrations automatically, but the principle of operation is always the same.

The phase shift $\Delta\phi$ that the plasma causes can be calculated as follows. If $k_0 = \omega/c$ is the propagation constant in air and k_1 is that in the plasma, we have

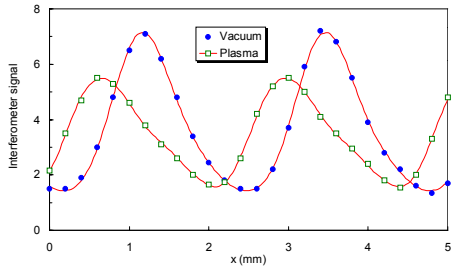


Fig. 5. Fringe shifts as the path length is changed. The lines are analytic fits through the points.

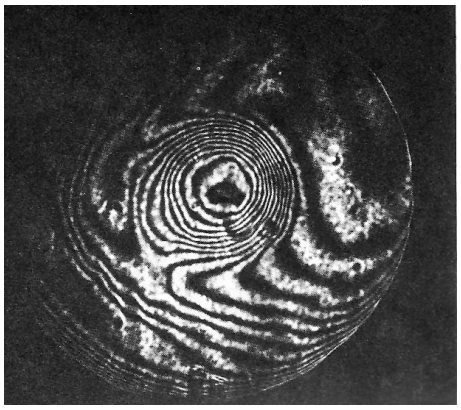


Fig. 6. Fringe patterns views along the axis can show the shape of the plasma [Heald and Wharton, 1978].

$$\Delta\phi = \int (k_0 - k_1)dx\,, \tag{2}$$

where k_1 is given by Eq. (1) as

$$k_1 = k_0\left(1 - \frac{n}{n_c}\right)^{1/2}. \tag{3}$$

Here we have replaced ω_p^2/ω^2 by n / n_c, where n_c is the "critical density" defined by

$$\omega^2 = \frac{n_c e^2}{\varepsilon_0 m}. \tag{4}$$

The phase shift is then

$$\Delta\phi = k_0 \int \left[1 - \left(1 - \frac{n(x)}{n_c}\right)^{1/2}\right] dx\,. \tag{5}$$

We see that the phase shift measures only the *line integral* of the density, not the local density. If ω is high enough that $n << n_c$, Eq. (5) can be Taylor expanded to obtain

$$\Delta\phi \approx \frac{k_0}{2n_c}\int n(x)dx \approx \tfrac{1}{2} k_0 L \frac{< n >}{n_c} \text{ radians}, \tag{6}$$

where $<n>$ is the average density over the path length L. In the reflection method, the integral (or L) must be doubled. Fig. 5 gives an example of the interferometer output in the double-pass method as the mirror is moved to change the path length. The fringe shift is clearly seen, but it is also evident that the waveform has been distorted. This is because the microwave generator did not give a pure signal, and its harmonics at higher ω suffered a different phase shift. By fitting the curves to sine waves and their harmonics and adjusting the relative phases, one can recover the phase shift of the fundamental and thus get the density. Fig. 6 shows an end view of a dense plasma, in which the path length was so long that many fringes are seen, revealing the shape of the plasma.

Microwave interferometry is useful for calibrating Langmuir probes. With a probe, one can measure the density profile across a radius or diameter of the plasma, but the absolute value of the density may not be known accurately. By using the measured density profile to compute the integral in Eq. (6), one can find the absolute density by measuring the microwave phase shift. The errors in this method come from the fact that the plasma is not perfectly planar, and the microwave beam is not

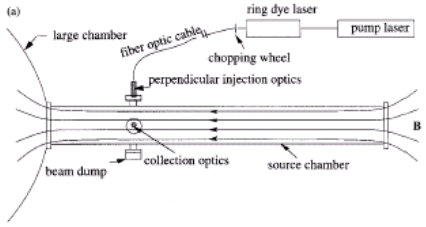


Fig. 7. Perpendicular alignment of injection laser and collection optics [Scime et al., Plasma Sources Sci. Technol. **7**, 186 (1998)].

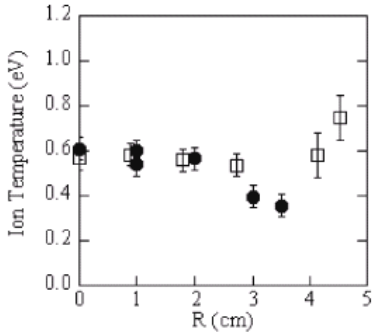


Fig. 8. LIF data on T_i parallel (solid points) and perpendicular (open points) to $\mathbf{B}_0$, showing anomalously high KT_i [Kline et al., Phys. Rev. Lett. **88**, 195002 (2002]).

perfectly parallel. Refraction can cause part of the beam to miss the collector, and reflections from the chamber walls can cause spurious waves.

3. Laser Induced Fluorescence (LIF)

This diagnostic is both non-invasive and local because it uses intersecting beam paths. Furthermore, it is the only way to measure T_i without using a large energy analyzer. One laser, tuned to a particular transition, is used to raise ions to an excited state along one path through the plasma. The excited ions fluoresce, giving off light at another frequency, and this light is collected by a lens focused to one part of the path, providing the localization. Doppler broadening of the line yields the ion velocity spread in a particular direction. The equipment is large, expensive, and difficult to set up, so that it is available in a relatively few laboratories. LIF is treated in more detail in Part B. Figure 7 shows a typical LIF setup, and Fig. 8 shows data taken in a helicon plasma.

XII. LANGMUIR PROBES

1. Construction and circuit

A Langmuir probe is small conductor that can be introduced into a plasma to collect ion or electron currents that flow to it in response to different voltages. The current vs. voltage trace, called the *I-V characteristic*, can be analyzed to reveal information about n, T_e, V_s (space potential), and even the distribution function $f_e(v)$, *but not the ion temperature*. Since the probe is immersed in a harsh environment, special techniques are used to protect it from the plasma and vice versa, and to ensure that the circuitry gives the correct $I-V$ values. The probe tip is made of a high-temperature material, usually a tungsten rod or wire 0.1–1 mm in diameter. The rod is threaded into a thin ceramic tube, usually alumina, to insulate it from the plasma except for a short length of exposed tip, about 2–10 mm long. These materials can be exposed to low-temperature laboratory plasmas without melting or excessive sputtering. To avoid disturbing the plasma, the ceramic tube should be as thin as possible, preferably < 1 mm in diameter but usually several times this. The probe tip should extend out of the end of the tube without touching it, so that it would not be in electrical contact with any conducting coating that may deposit onto the insulator. The assembly is encased in a vacuum jacket, which could be a stainless steel or glass tube 1/4″ in outside diameter (OD). It is preferable to make the vacuum seal at the outside end of the probe as-

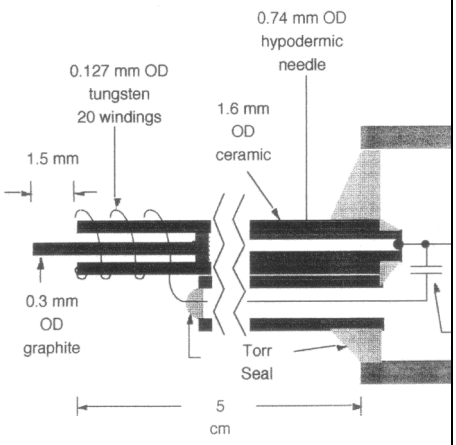


Fig. 9. A carbon probe tip assembly with RF compensation circuitry [Sudit and Chen, Plasma Sources Sci. Technol. **4**, 162 (1994)].

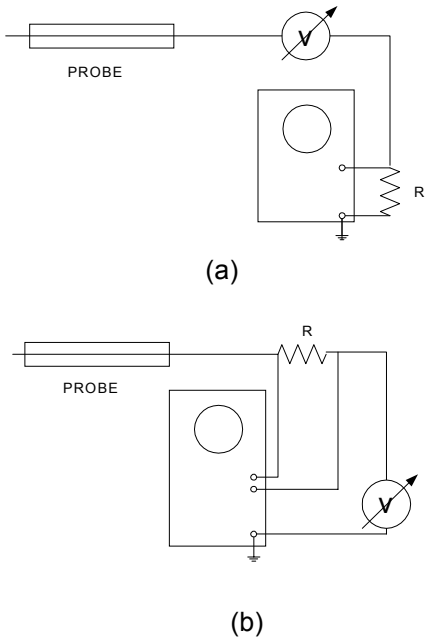


Fig. 10. Two basic configurations for the probe circuit.

sembly rather than at the end immersed in the plasma, which can cause a leak. Only the ceramic part of the housing should be allowed to enter the plasma. Some commercial Langmuir probes use a rather thick metal tube to support the probe tip assembly, and this can modify the plasma characteristics unless the density is very low. In dense plasmas the probe cannot withstand the heat unless the plasma is pulsed or the probe is mechanically moved in and out of the plasma in less than a second. When collecting ion current, the probe can be eroded by sputtering, thus changing its collection area. This can be minimized by using carbon as the tip material. Ordinary pencil lead, 0.3mm in diameter works well and can be supported by a hypodermic needle inside the ceramic shield. One implementation of a probe tip assembly is shown in Fig. 9.

There are two basic ways to apply a voltage V to the probe and measure the current I that it draws from the plasma, and each has its disadvantages. In Fig. 10a, the probe lead, taken through a vacuum feedthru, is connected to a battery or a variable voltage source (*bias supply*) and then to a termination resistor R to ground. To measure the probe current, the voltage across R is recorded or displayed on an oscilloscope. This arrangement has the advantage that the measuring resistor is grounded and therefore not subject to spurious pickup. Since the resistor is usually 10-1000Ω, typically 50Ω, this is not a serious problem anyway. The disadvantage is that the bias supply is floating. If this is a small battery, it cannot easily be varied. If it is a large electronic supply, the capacitance to ground will be so large that ac signals will be short-circuited to ground, and the probe cannot be expected to have good frequency response. The bias supply can also act as an antenna to pick up rf noise. To avoid this, one can ground the bias supply and put the measuring resistor on the hot side, as shown in Fig. 10b. This is usually done if the bias supply generates a sweep voltage. However, the voltage across R now has to be measured with a differential amplifier or some other floating device; or, it can be optoelectronically transmitted to a grounded circuit. The probe voltage V_p should be measured on the ground side of R so as not to load the probe with another stray capacitance.

To measure plasma potential with a Langmuir probe, one can terminate the probe in a high impedance, such as the 1 MΩ input resistance of the oscilloscope. This is called a *floating probe*. A lower R, like 100K, can be used to suppress pickup. The minimum value of

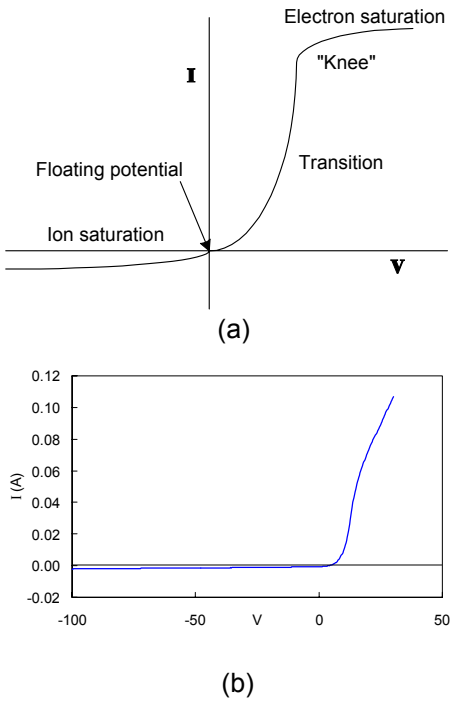


Fig. 11. (a) An idealized *I* – *V* characteristic showing its various parts; (b) a real *I* – *V* curve from an ICP.

R has to be high enough that the *IR* drop through it does change the measured voltage. A rough rule of thumb is that $I_{sat}R$ should be much greater than T_{eV}, or $R >> T_{eV}/I_{sat}$, where I_{sat} is the ion saturation current defined below. The voltage measured is not the plasma potential but the floating potential, also defined below. The large value of *R* means that good frequency response is difficult to achieve because of the *RC* time constant of stray capacitances. One can improve the frequency response with *capacitance neutralization* techniques, but even then it is hard to make a floating probe respond to RF frequencies.

2. The electron characteristic

A Langmuir *I-V* trace is usually displayed upside down, so that electron current into the probe is in the +*y* direction. The curve, resembling that in Fig. 11a, has five distinct parts. The point at which the curve crosses the *V* axis is called the *floating potential* V_f. To the left of this the probe draws ion current, and the curve soon flattens out to a more or less constant value called the *ion saturation current* I_{sat}. To the right of V_f, electron current is drawn, and the *I-V* curve goes into an *exponential part*, or *transition region*, as the Coulomb barrier is lowered to allow slower electrons in the Maxwellian distribution to penetrate it. At the space potential V_s, the curve takes a sharp turn, called the *knee*, and saturates at the *electron saturation current* I_{es}. Actual $I - V$ curves in RF or magnetized plasmas usually have an indistinct knee, as shown in Fig. 11b.

The exponential part of the $I - V$ curve, when plotted semi-logarithmically vs. the probe voltage V_p, should be a straight line if the electrons are Maxwellian:

$$I_e = I_{es} \exp[e(V_p - V_s)/KT_e)], \qquad (7)$$

where, from Eq. (A4-2),

$$I_{es} = eAn_e\bar{v}/4 = en_eA\left(\frac{KT_e}{2\pi m}\right)^{1/2}, \qquad (8)$$

A being the exposed area of the probe tip. Eq. (7) shows that the slope of the $(\ln I)$–V_p curve is exactly $1/T_{eV}$ and is a good measure of the electron temperature. As long as the electrons are Maxwellian and are repelled by the probe, the EEDF at a potential $V < 0$ is proportional to

$$f(\mathrm{v}) \propto \mathrm{e}^{-(\frac{1}{2}m\mathrm{v}^2 + eV)/KT_e} = \mathrm{e}^{-e|V|/KT_e}\mathrm{e}^{-(m\mathrm{v}^2/2KT_e)}. \qquad (9)$$

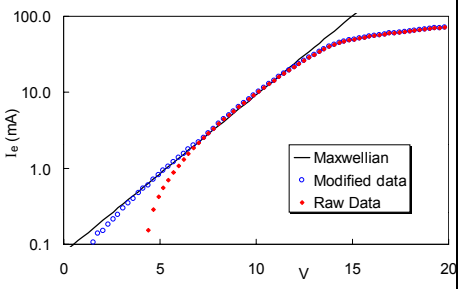


Fig. 12. A semilog plot of electron current from an *I* – *V* curve in an ICP.

We see that $f(v)$ is still Maxwellian at the same T_e; only the density is decreased by $\exp(-e|V|/KT_e)$. Thus, the slope of the semilog curve is independent of probe area or shape and independent of collisions, since these merely preserve the Maxwellian distribution. However, before I_e can be obtained from I, one has to subtract the ion current I_i. This can be done approximately by drawing a straight line through I_{sat} and extrapolating it to the electron region.

One can estimate the ion contribution more accurately by using one of the theories of ion collection discussed below, but refinements to this small correction are usually not necessary, and they affect only the high-energy tail of the electron distribution. One easy iteration is to change the magnitude of the I_{sat} correction until the ln I plot is linear over as large a voltage range as possible. Fig. 12 shows a measured electron characteristic and a straight-line fit to it. The ion current was calculated from a theoretical fit to I_{sat} and added back to I to get I_e. The uncorrected points are also shown; they have a smaller region of linearity.

3. Electron saturation

Since I_{sat} is $\approx c_s$ because of the Bohm sheath criterion, I_{es}, given by Eq. (8), should be $\approx(M/m)^{1/2}$ times as large as I_{sat}. In low-pressure, unmagnetized discharges, this is indeed true, and the knee of the curve is sharp and is a good measure of V_s. For very high positive voltages, I_{es} increases as the sheath expands, the shape of the curve depending on the shape of the probe tip. However, effects such as collisions and magnetic fields will lower the magnitude of I_{es} and round off the knee so that V_s is hard to determine. In particular, magnetic fields strong enough to make the electron Larmor radius smaller than the probe radius will limit I_{es} to only 10-20 times I_{sat} because the probe depletes the field lines that it intercepts, and further electrons can be collected only if they diffuse across the B-field. The knee, now indistinct, indicates a space potential, but only that in the depleted tube of field lines, not V_s in the main plasma. In this case, the $I - V$ curve is exponential only over a range of a few KT_e above the floating potential and therefore samples only the electrons in the tail of the Maxwellian. One might think that measurement of I_{es} would give information on the electron density, but this is possible only at low densities and pressures, where the mean free path is very long. Otherwise, the current collected by the probe is so large that it drains the plasma and changes its equilibrium

properties. It is better to measure n by collecting ions, which would give the same information, since plasmas are quasineutral. More importantly, one should avoid collecting saturation electron current for more than a few milliseconds at a time, because the probe can be damaged.

4. Space potential

The time-honored way to obtain the space potential (or plasma potential) is to draw straight lines through the $I - V$ curve in the transition and electron saturation regions and call the crossing point V_s, I_{es}. This does not work well if I_{es} region is curved. As seen in Fig. 11b, a good knee is not always obtained even in an ICP with $\mathbf{B}_0 = 0$. In that case, there are two methods one can use. The first is to measure V_f and calculate it from Eq. (A4-4), regarding the probe as a wall. The second is to take the point where I_e starts to deviate from exponential growth; that is, where $I_e'(V)$ is maximum or $I_e''(V)$ is zero. If $I_e'(V)$ has a distinct maximum, a reasonable value for V_s is obtained, but it would be dangerous to equate the current there to I_{es}. That is because, according to Eq. (7), I_{es} depends exponentially on the assumed value of V_s.

5. Ion saturation current[1]

a) Plane probes. The measurement of I_{sat} is the simplest and best way to determine n. At densities above about 10^{11} cm^{-3}, the sheath around a negatively biased probe is so thin that the area of the sheath edge is essentially the same as the area of the probe tip itself. The ion current is then just that necessary to satisfy the Bohm sheath criterion:

$$I_{sat} = 0.5eAn(KT_e / M)^{1/2}, \qquad (10)$$

where the factor 0.5 represents n_s/n. This value is only approximate; when probes are calibrated against other diagnostics, such as microwave interferometry, a factor of 0.6-0.7 has been found to be more accurate. Note that Eq. (10) predicts a constant I_{sat}, which can happen only for flat probes in which the sheath area cannot expand as the probe is made more and more negative. In practice, I_{sat} usually has a slope to it. This is because the ion current has to come from a disturbed volume of plasma (the presheath) where the ion distribution changes from iso-

[1] For detailed references, see F.F. Chen, *Electric Probes*, in "Plasma Diagnostic Techniques", ed. by R.H. Huddlestone and S.L. Leonard (Academic Press, New York, 1965), Chap. 4, pp. 113-200.

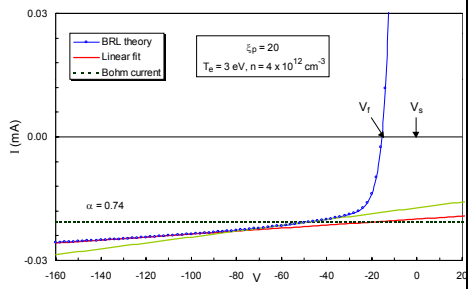


Fig. 13. Illustrating the extrapolation of I_i back to the floating potential to get I_{sat}. In this case, the Bohm coefficient 0.5 in Eq. (10) has to be replaced by 0.74 to get the right density.

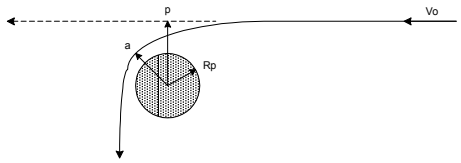


Fig. 14. Definition of impact parameter *p*.

tropic to unidirectional. If the probe is a disk of radius R, say, the disturbed volume may have a size comparable to R, and would increase as the $|V_p|$ increases. In that case, one can extrapolate I_i back to V_f to get a better measure of I_{sat} before the expansion of the presheath. This is illustrated in Fig. 13. Better saturation with a plane probe can be obtained by using a *guard ring*, a flat washer-shaped disk surrounding the probe but not touching it. It is biased at the same potential as the probe to keep the fields planar as V_p is varied. The current to the guard ring is disregarded. A section of the chamber wall can be isolated to be used as a plane probe with a large guard ring.

b) Cylindrical probes

i) OML theory. As the negative bias on a probe is increased to draw I_i, the sheath on cylindrical and spherical probes expands, and I_i does not saturate. Fortunately, the sheath fields fall off rapidly away from the probe so that exact solutions for $I_i(V_p)$ can be found. We consider cylindrical probes here because spherical ones are impractical to make, though the theory for them converges better. The simplest theory is the orbital-motion-limited (OML) theory of Langmuir.

Consider ions coming to the attracting probe from infinity in one direction with velocity v_0 and various impact parameters p. The plasma potential V is 0 at ∞ and is negative everywhere, varying gently toward the negative probe potential V_p . Conservation of energy and angular momentum give

$$\begin{aligned} &½\, mv_0^2 = ½\, mv_a^2 + eV_a \equiv -eV_0 \\ &pv_0 = av_a \end{aligned} \qquad (11)$$

where $eV < 0$ and a is the distance of closest approach to the probe of radius R_p . Solving, we obtain

$$½mv_a^2 = ½mv_0^2\left(1+\frac{V_a}{V_0}\right), \qquad p = a\frac{v_a}{v_0} = a\left(1+\frac{V_a}{V_0}\right)^{1/2} \quad (12)$$

If $a \le R_p$, the ion is collected; thus, the *effective* probe radius is $p(R_p)$. For monoenergetic particles, the flux to a

[2] F.F. Chen, Phys. Plasmas **8**, 3029 (2001).
[3] F.F. Chen, J.D. Evans, and D. Arnush, Phys. Plasmas **9**, 1449 (2002)
[4] I.D. Sudit and F.F. Chen, *RF compensated probes for high-density discharges*, Plasma Sources Sci. Technol. **3**, 162 (1994).
[5] N. Hershkowitz, *How Langmuir Probes Work,* in Plasma Diagnostics, Vol. 1, Ed. by O. Auciello and D.L. Flamm (Acad. Press, N.Y., 1994), Chap. 3, p. 113.

probe of length L is therefore

$$\Gamma = 2\pi R_p L(1 + V_a / V_0)^{1/2}\Gamma_r , \qquad (13)$$

where Γ_r is the random flux of ions of that energy. Langmuir then extended this result to energy distributions which were Maxwellian at some large distance $r = s$ from the probe, where s is the "sheath edge". The random flux Γ_r is then given by the usual formula

$$\Gamma_r = n\left(\frac{KT_i}{2\pi M}\right)^{1/2} . \qquad (14)$$

With A_p defined as the probe area, integrating over all velocities yields the cumbersome expression

$$\Gamma = A_p\Gamma_r\left\{\frac{s}{a}\mathrm{erf}(\Phi^{1/2}) + e^{\chi}[1 - \mathrm{erf}(\chi + \Phi)^{1/2}]\right\} , \qquad (15)$$

where $\chi \equiv -eV_p / KT_i$, $\Phi \equiv \left(\frac{a^2}{s^2 - a^2}\right)\chi$, $a = R_p$.

Fortunately, there are small factors. In the limit $s >> a$, when OML theory applies, if at all, we have $\Phi << \chi$, and for $T_i \to 0$, $1/\chi << 1$. Expanding in Taylor series, we find that the T_i dependences of χ and Γ_r cancel, and a finite limiting value of the OML current exists, independently of the value of T_i.

$$I \xrightarrow[T_i \to 0]{} A_p ne\frac{\sqrt{2}}{\pi}\left(\frac{|eV_p|}{M}\right)^{1/2} . \qquad (16)$$

Thus, the OML current is proportional to $|V_p|^{1/2}$, and the $I - V$ curve is a parabola, while the $I^2 - V$ curve is a straight line. This scaling is the result of conservation of energy and angular momentum. Because ions have large angular momentum at large distances, though they have small velocities, they tend to orbit the probe and miss it. The probe voltage draws them in. The value of T_i cancels out mathematically, but T_i has to be finite for this physical mechanism to work.

The OML result, though simple, is very restricted in applicability. Since the sheath radius s was taken to be infinite, the density has to be so low that the sheath is much larger than the probe. The potential variation $V(r)$ has to be gentle enough that there does not exist an "absorption radius" inside of which the E-field is so strong that no ions can escape being collected. Except in very tenuous plasmas, a well developed sheath and an absorp-

tion radius exist, and OML theory is inapplicable. Nonetheless, the $I^2 - V$ dependence of I_{sat} is often observed and is mistakenly taken as evidence of orbital motion.

ii) ABR theory. To do a proper sheath theory, one has to solve Poisson's equation for the potential $V(r)$ everywhere from the probe surface to $r = \infty$. Allen, Boyd, and Reynolds (ABR) simplified the problem by assuming *ab initio* that $T_i = 0$, so that there are no orbital motions at all: the ions are all drawn radially into the probe. Originally, the ABR theory was only for spherical probes, but it was later extended to cylindrical probes by Chen[1], as follows. Assume that the probe is centered at $r = 0$ and that the ions start at rest from $r = \infty$, where $V = 0$. Poisson's equation in cylindrical coordinates is

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial V}{\partial r}\right) = \frac{e}{\varepsilon_0}\left(n_e - n_i\right), \qquad n_e = n_0 e^{eV/KT_e}. \quad (17)$$

To electrons are assumed to be Maxwellian. To find n_i, let I be the total ion flux per unit length collected by the probe. By current continuity, the flux per unit length at any radius r is

$$\Gamma = n_i v_i = I/2\pi r, \quad \text{where} \quad v_i = \left(-2eV/M\right)^{1/2}. \quad (18)$$

$$\text{Thus,} \qquad n_i = \frac{\Gamma}{v_i} = \frac{I}{2\pi r}\left(\frac{-2eV}{M}\right)^{-1/2}. \quad (19)$$

Poisson's equation can then be written

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial V}{\partial r}\right) = -\frac{e}{\varepsilon_0}\left[\frac{I}{2\pi r}\left(\frac{-2eV}{M}\right)^{-1/2} - n_0 e^{eV/KT_e}\right] \quad (20)$$

Defining

$$\eta \equiv -\frac{eV}{KT_e}, \qquad c_s \equiv \left(\frac{KT_e}{M}\right)^{1/2}, \quad (21)$$

we can write this as

$$\frac{KT_e}{e}\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \eta}{\partial r}\right) = -\frac{e}{\varepsilon_0}\left[\frac{I}{2\pi r}\frac{(2\eta)^{-1/2}}{c_s} - n_0 e^{-\eta}\right] \quad (22)$$

or

$$-\frac{\varepsilon_0 KT_e}{n_0 e^2}\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \eta}{\partial r}\right) = \frac{I}{2\pi r}\frac{(2\eta)^{-1/2}}{n_0 c_s} - e^{-\eta}. \quad (23)$$

The Debye length appears on the left-hand side as the natural length for this equation. We therefore normalize r to λ_D by defining a new variable ξ:

$$\xi \equiv \frac{r}{\lambda_D}, \qquad \lambda_D \equiv \left(\frac{\varepsilon_0 K T_e}{n_0 e^2} \right)^{1/2}. \tag{24}$$

Eq. (23) now becomes

$$\begin{aligned} \frac{\partial}{\partial \xi}\left(\xi \frac{\partial \eta}{\partial \xi} \right) &= \frac{I\xi}{2\pi r} \frac{1}{n_0 c_s} (2\eta)^{-1/2} - \xi e^{-\eta} \\ &= \frac{I}{2\pi n_0} \frac{1}{\lambda_D c_s} (2\eta)^{-1/2} - \xi e^{-\eta} \\ &= \frac{I}{2\pi n_0} \left(\frac{n_0 e^2}{\varepsilon_0 K T_e} \frac{M}{K T_e} \right)^{1/2} (2\eta)^{-1/2} - \xi e^{-\eta} \\ &= \frac{eI}{2\pi K T_e} \left(\frac{M}{2\varepsilon_0 n_0} \right)^{1/2} \eta^{-1/2} - \xi e^{-\eta} \end{aligned} \tag{25}$$

Defining

$$J \equiv \frac{eI}{2\pi K T_e} \left(\frac{M}{2\varepsilon_0 n_0} \right)^{1/2}, \tag{26}$$

we arrive at the ABR equation for cylindrical probes:

$$\frac{\partial}{\partial \xi}\left(\xi \frac{\partial \eta}{\partial \xi} \right) = J \eta^{-1/2} - \xi e^{-\eta}. \tag{27}$$

For each assumed value of J (normalized probe current), this equation can be integrated from $\xi = \infty$ to any arbitrarily small ξ. The point on the curve where $\xi = \xi_p$ (the probe radius) gives the probe potential η_p for that value of J. By computing a family of curves for different J (Fig. 15), one can obtain a $J - \eta_p$ curve for a probe of radius ξ_p by cross-plotting (Fig 16). Of course, both J and ξ_p depend on the unknown density n_0, which one is trying to determine from the measured current I_i. (KT_e is supposed to be known from the electron characteristic.) The extraction of n_0 from these universal curves is a trivial matter for a computer. In the graphs the quantity $J\xi_p$ is plotted, since that is independent of n_0. Note that for small values of ξ_p, I^2 varies linearly with V_p, as in OML theory, but for entirely different reasons, since there is no orbiting here.

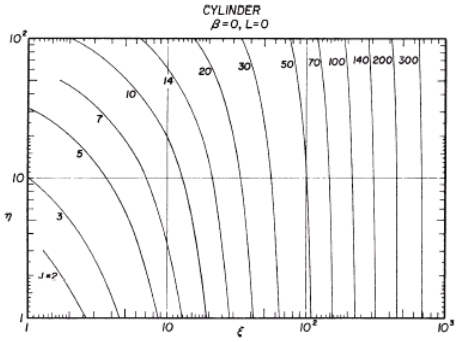


Fig. 15. ABR curves for $\eta(\xi)$.

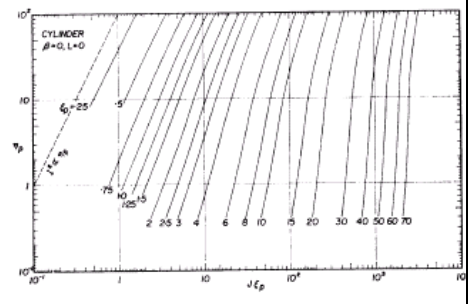


Fig. 16. *V–I* curves derived from $\eta(\xi)$.

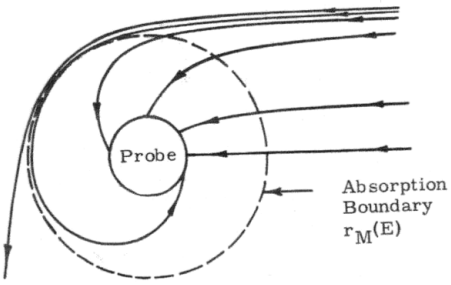


Fig. 17. Definition of absorption radius.

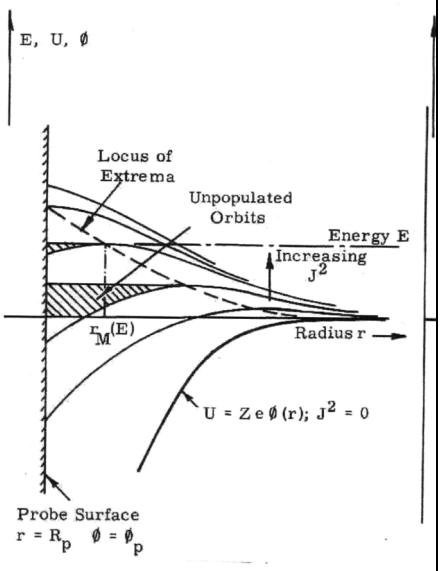


Fig. 18. Effective potential seen by ions with angular momentum J.

iii) BRL theory. The first probe theory which accounted for both sheath formation and orbital motions was published by Bernstein and Rabinowitz (BR), who assumed an isotropic distribution of ions of a single energy E_i. This was further refined by Laframboise (L), who extended the calculations to a Maxwellian ion distribution at temperature T_i. The BRL treatment is considerably more complicated than the ABR theory. In ABR, all ions strike the probe, so the flux at any radius depends on the conditions at infinity, regardless of the probe radius. That is why there is a set of universal curves. In BRL theory, however, the probe radius must be specified beforehand, since those ions that orbit the probe will contribute twice to the ion density at any given radius r, while those that are collected contribute only once. The ion density must be known before Poisson's equation can be solved, and clearly this depends on the presence of the probe. There is an "absorption radius" (Fig. 17), depending on J, inside of which all ions are collected. Bernstein solved the problem by expressing the ion distribution in terms of energy E and angular momentum J instead of v_r and $v_\perp$. Ions with a given J see an effective potential barrier between them and the probe. They must have enough energy to surmount this barrier before they can be collected. In Fig. 18, the lowest curve is for ions with $J = 0$; these simply fall into the probe. Ions with finite J see a potential hill. With sufficient energy, they can climb the hill and fall to the probe on the other side. The dashed line through the maxima shows the absorption radius for various values of J.

The computation tricky and tedious. It turns out that KT_i makes little difference if $T_i/T_e < 0.1$ or so, as it usually is. Laframboise's extension to a Maxwellian ion distribution is not normally necessary; nonetheless, Laframboise gives the most complete results. Fig. 19 shows an example of ion saturation curves from the BRL theory. One sees that for large probes ($R_p/\lambda_D >> 1$) the ion current saturates well, since the sheath is thin. For small R_p/λ_D, I_i grows with increasing V_p as the sheath radius increases.

One might think that the ABR result would be recovered if takes $T_i = 0$ or $E_i = 0$ in the BRL computation. However, this happens only for spherical probes. For cylindrical probes, there is a problem of nonuniform convergence. Since the angular momentum is Mvr, for $r \to \infty$ ions with zero thermal velocity have $J = (M)(0)(\infty)$, an indeterminate form. The correct treatment is to calculate the probe current for $T_i > 0$ and *then* take the limit T_i

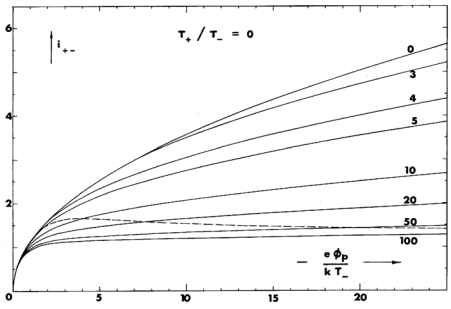


Fig. 19. Laframboise curves for $I_i - V$ characteristics in dimensionless units, in the limit of cold ions. Each curve is for a different ratio R_p/λ_D.

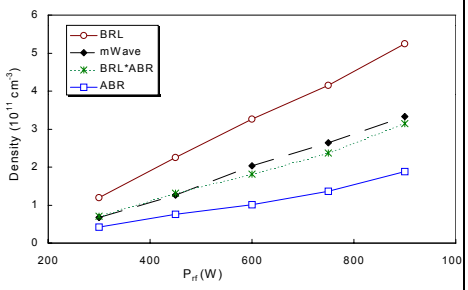


Fig. 20. Comparison of n measured with microwaves with probes using two different probe theories.

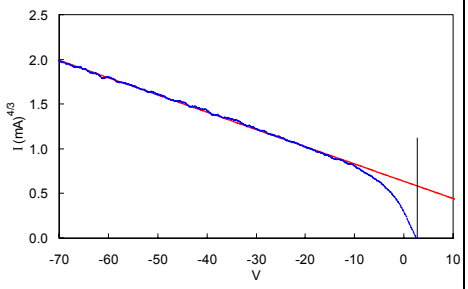


Fig. 21. Extrapolation to get I_i at V_f.

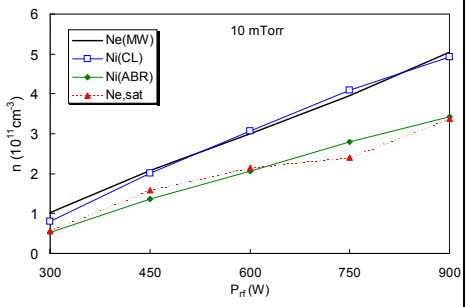


Fig. 22. Comparison of microwave and probe densities using the floating potential method (CL), ABR theory, and $I_{e,sat}$.

$\rightarrow 0$, as BRL have done. The BRL predictions have been borne out in experiments in fully ionized plasmas, but not in partially ionized ones.

iv) Comparison among theories. It is not reasonable to reproduce the ABR or BRL computations each time one makes a probe measurement. Chen[2] has solved this problem by parametrizing the ABR and BRL curves so that the $I_i - V$ curve can be easily be created for any value of R_p/λ_D. One can then compute the plasma density from the probe data using the ABR and BRL theories and compare with the density measured with microwave interferometry. Such a comparison is shown in Fig. 20. One sees that the ABR theory predicts too low a density because orbiting is neglected, and therefore the predicted current is too high and the measured current is identified with a lower density. Conversely, BRL theory predicts too high a density because it assumes more orbiting than actually occurs, so that the measured current is identified with a high density. This effect occurs in partially ionized plasmas because the ions suffer charge-exchange collisions far from the probe, outside the sheath, thus losing their angular momentum. The BRL theory assumes that the ions retain their angular momentum all the way in from infinity. One might expect the real density to lie in between, and indeed, it agrees quite well with the geometric mean of the BRL and ABR densities.

Treating the charge-exchange collisions rigorously in the presheath would be an immense problem, but recently Chen et al.[3] have found an even easier, fortuitous, way to estimate the plasma density in ICPs and other processing discharges. The method relies on finding the ion current at floating potential V_f by extrapolating on a graph of $I_i^{4/3}$ vs. V_p, as shown in Fig. 21. The power 4/3 is chosen because it usually leads to a straight line graph. At $V_p = V_f$, let the sheath thickness d be given by the Child-Langmuir formula of Eq. (A4-7) with $V_0 = V_f$. The sheath area is then $A = 2\pi(R_p+d)L$. If the ions enter the sheath at velocity c_s with density $n_s = \frac{1}{2}n_0c_s$, the ion current is $I_i = \frac{1}{2}n_0eAc_s$, and n can easily be calculated from the extrapolated value of $I_i(V_f)$. Note that if $R_p << d$, Eq. (A4-7) predicts the $I_i^{4/3} - V_p$ dependence (but this is accidental). (Since $d \propto \lambda_D \propto n^{-1/2}$, n has to be found by iteration or by solving a quadratic equation.) The density in a 10 mTorr ICP discharge is shown in Fig. 22, compared to n measured by microwaves, by probes using the ABR theory, or by probes using the saturation electron current. The V_f(CL) method fits best, though the fit

is not always this good. The OML theory (not shown) also fits poorly. Though this is a fast and easy method to interpret I_{sat} curves, it is hard to justify because the CL formula of Eq. (A4-7) applies to planes, not cylinders, and the Debye sheath thickness has been neglected, as well as orbiting and collisions. This simple-minded approach apparently works because the neglected effects cancel one another. From the preceding discussion, it is clear that the rigorous theories, ABR and BRL, can err by a factor of 2 or more in the value of n in partially ionized plasmas. There are heuristic methods, but these may not work in all conditions. It is difficult for Langmuir probes to give a value of n accurate to better than 10–20%; fortunately, such accuracy is not often required.

6. Distribution functions

Since the ion current is insensitive to T_i, Langmuir probes cannot measure ion temperature, and certainly not the ion velocity distribution. However, careful measurement of the transition region of the $I - V$ characteristic can reveal the electron distribution if it is isotropic. If the probe surface is a plane perpendicular to x, the electron flux entering the sheath depends only on the x component of velocity, v_x. For instance, the Maxwell distribution for v_x is

$$f_M(\mathrm{v}_x) = \left(\frac{1}{\mathrm{v}_{th}\sqrt{\pi}}\right)\exp(-\mathrm{v}_x^2/\mathrm{v}_{th}^2), \quad \mathrm{v}_{th}^2 \equiv 2KT_e/m\,. \tag{28}$$

The coefficient normalizes $f(\mathrm{v})$ so that its integral over all v_x's is unity. If $f(\mathrm{v})$ is not Maxwellian, it will have another form and another coefficient in front. The electron current that can get over the Coulomb barrier and be collected by the probe will therefore be

$$I_e = eAn\int_{\mathrm{v}_{\min}}^{\infty} \mathrm{v}_x f(\mathrm{v}_x)d\mathrm{v}_x\ ,\ \tfrac{1}{2}m\mathrm{v}_{\min}^2 = e(V_s - V_p) = -eV_p \tag{29}$$

where $\mathrm{v}_{\min}$ is the minimum energy of an electron that can reach the probe, and $V_s = 0$ by definition. Taking the derivative and simplifying, we find

$$\frac{dI_e}{dV_p} = eAn\frac{d}{dV_p}\int_{\mathrm{v}_{\min}}^{\infty} \mathrm{v}_x f(\mathrm{v}_x)\frac{d\mathrm{v}_x}{dV_p}dV_p$$

$$= -eAn\,\mathrm{v}_x f(\mathrm{v}_x)\frac{d\mathrm{v}_x}{dV_p}\bigg|_{\mathrm{v}_x=\mathrm{v}_{\min}}$$

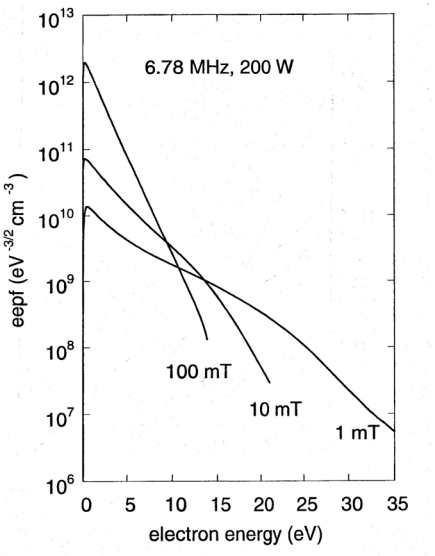


Fig. 23. EEDF curves obtained with a Langmuir probe in a TCP discharge [Godyak et al. J. Appl. Phys. **85**, 3081, (1999)].

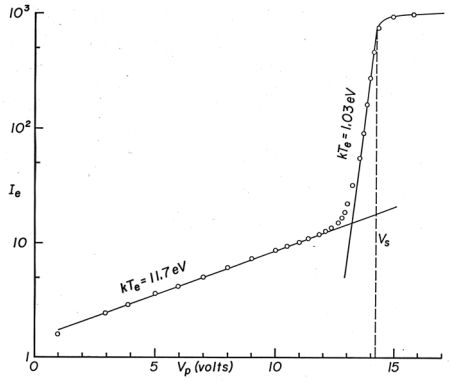
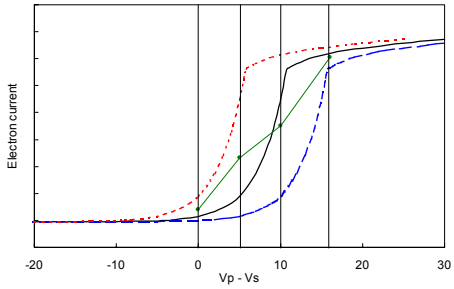
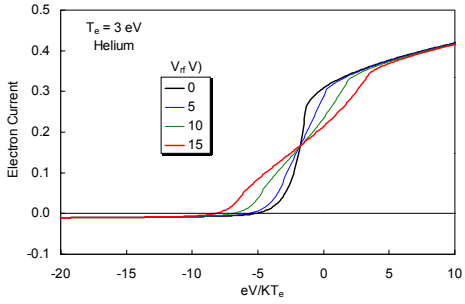


Fig. 24. An $I - V$ curve of a bi-Maxwellian EEDF.



(a)



(b)

Fig. 25 . (a) The center curve is the correct $I - V$ curve. The dashed ones are displaced by ±5V, representing changes in V_s. At the vertical lines, the average I_e between the displaced curves is shown by the dot. The line through the dots is the time-averaged $I - V$ curve that would be observed, differing greatly from the correct curve. (b) Computed $I - V$ curves for sinusoidal V_s oscillations of various amplitudes.

$$\frac{dI_e}{dV_p} = -eAn\,\mathrm{v}_x f(\mathrm{v}_x)\frac{-e}{m\mathrm{v}_x}\bigg|_{\mathrm{v}_x=\mathrm{v}_{\min}} = A\frac{ne^2}{m} f(\mathrm{v}_{\min}), \quad (30)$$

so that $f(\mathrm{v}_x)$ can be found from the first derivative of the $I - V$ curve. If the probe is not flat, however, one has to take the three-dimensional distribution $g(v) = 4\pi \mathrm{v}^2 f(\mathrm{v})$, where v is the absolute value |v| of the velocity, and take into account the various angles if incidence. Without going into the details, we then find, surprisingly, that $f(\mathrm{v})$ is proportional to the *second* derivative of the $I - V$ curve:

$$\frac{d^2 I_e}{dV_p^2} \propto f(\mathrm{v}) \quad (31)$$

This result is valid for any convex probe shape as long as the distribution is isotropic, and for any anisotropic distribution if the probe is spherical. To differentiate $I - V$ data twice will yield noisy results unless a good deal of smoothing is employed. Alternatively, one can *dither* the probe voltage by modulating it at a low frequency, and the signal at the dither frequency will be proportional to the first derivative. In that case, only one further derivative has to be taken to get $f(\mathrm{v})$. Figure 23 is an example of non-Maxwellian $f(\mathrm{v})$'s obtained by double differentiation with digital filtering.

In special cases where the EEDF consists of two Maxwellians with well separated temperatures, the two KT_e's can be obtained by straight-line fits on the semilog $I - V$ curve without complicated analysis. An example of this is shown in Fig. 24.

7. RF compensation

Langmuir probes used in RF plasma sources are subject to RF pickup which can greatly distort the $I - V$ characteristic and give erroneous results. ECR sources which operate in the microwave regime do not have this trouble because the frequency is so high that it is completely decoupled from the circuitry, and the measured currents are the same as in a DC discharge. However, in RF plasmas, the space potential can fluctuate is such a way that the circuitry responds incorrectly. The problem is that the $I - V$ characteristic is nonlinear. The "V" is actually the potential difference $V_p - V_s$, where V_p is a DC potential applied to the probe, and V_s is a potential that can fluctuate at the RF frequency and its harmonics. If one displaces the $I - V$ curve horizontally back and

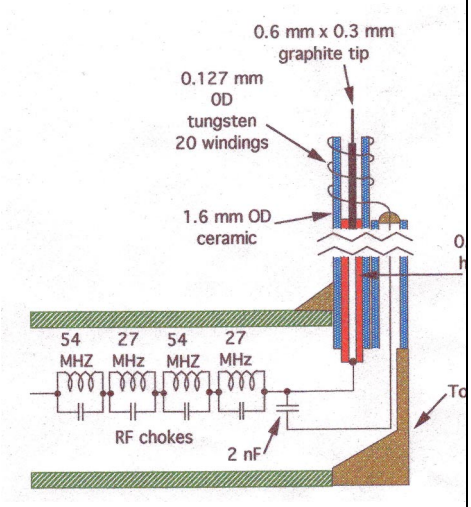


Fig. 26. Design of a dogleg probe.

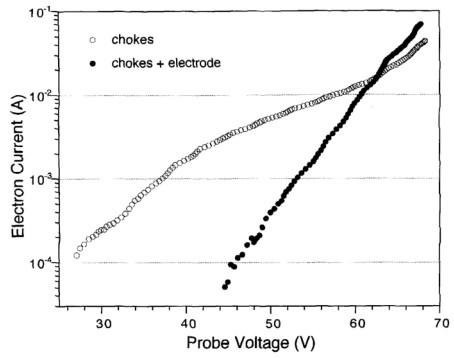


Fig. 27. *I* – *V* curves taken with and without an auxiliary electrode.

forth around a center value V_0, the average current I measured will not be $I(V_0)$, since I varies exponentially in the transition region and also changes slope rapidly as it enters the ion and electron saturation regions. The effect of this is to make the $I - V$ curve wider, leading to a falsely high value of T_e and shifting the floating potential V_f to a more negative value. This is illustrated in Fig. 25.

Several methods are available to correct for this. One is to tap off a sinusoidal RF signal from the power supply and mix this with the probe signal with variable phase and amplitude. When the resultant $I - V$ curve gives the lowest value of T_e, one has probably simulated the V_s oscillations. This method has the disadvantage that the V_s oscillations can contain more than one harmonic. A second method is to measure the V_s oscillations with another probe or section of the wall which is floating, and add that signal to the probe current signal with variable phase and amplitude. The problem with this method is that the V_s fluctuations are generally not the same everywhere. A third method is to isolate the probe tip from the rest of the circuit with an RF choke (inductor), so that the probe tip is floating at RF frequencies but is fixed at the DC probe bias at low frequencies. The problem is that the probe tip does not draw enough current to fill the stray capacitances that connect it to ground at RF frequencies. One way is to place a large slug of metal inside the insulator between the probe tip and the chokes. This metal slug has a large area and therefore picks up enough charge from the V_s oscillations to drive the probe tip to follow them. However, we have found[4] that the best way is to use an external floating electrode, which could be a few turns of wire around the probe insulator, and connect it through a capacitor to a point between the probe tip and the chokes. The charge collected by this comparatively large "probe" is then sufficient to drive the probe tip so that V_p - V_s remains constant. Note that this auxiliary electrode supplies only the RF voltage; the dc part is still supplied by the external power supply. The design of the chokes is also critical: they must have high enough Q to present a resonantly high impedance at both the fundamental and the second harmonic of the RF frequency. This is the reason there are two pairs of chokes in Fig. 9. One pair is resonant at ω, and the other at 2ω. Two chokes are used in series to increase the Q. A compromise has to be made between high Q and small physical size of the chokes. Figure 26 shows a "dogleg" design, which permits scans in two directions. Figure 27 shows an $I - V$ curve taken with and without the auxiliary electrode, showing that the chokes

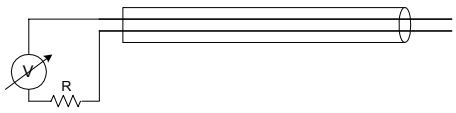


Fig. 28. A double probe.



Fig. 29. A hot probe.

themselves are usually insufficient.

Without proper RF compensation, Langmuir probe data in RF discharges can give spurious data on T_e, V_f, and $f(v)$. However, if one needs to find only the plasma density, the probe can be biased so that V never leaves the ion saturation region, which is linear enough that the average I_{sat} will be the correct value.

8. Double probes and hot probes[5]

When V_s fluctuates slowly, one can use the method of double probes, in which two identical probes are inserted into the plasma in close proximity, and the current from one to the other is measured as a function of the voltage difference between them. The $I - V$ characteristic is then symmetrical and limited to the region between the I_{sat}'s on each probe. If the probe array floats up and down with the RF oscillations, the $I - V$ curve should not be distorted. This method does not work well in RF plasmas because it is almost impossible to make the whole two-probe system float at RF frequencies because of the large stray capacitance to ground. Even if both tips are RF compensated, the RF impedances must be identical.

Hot probes are small filaments that can be heated to emit electrons. These electrons, which have very low energies corresponding to the KT of the filament, cannot leave the probe as long as $V_p - V_s$ is positive. As soon as $V_p - V_s$ goes negative, however, the thermionic current leaves the probe, and the probe current is dominated by this rather than by the ion current. Where the $I - V$ curve crosses the x axis, therefore, is a good measure of V_s. The voltage applied to the filament to heat it can be eliminated by turning it off and taking the probe data before the filament cools. One can also heat the probe by bombarding it with ions at a very large negative V_p, and then switching this voltage off before the measurement. In general it is tricky to make hot probes small enough. For further information on these techniques and on behavior of Langmuir probes in RF plasmas, the reader is referred to the chapter by Hershkowitz[5].

XIII. OTHER LOCAL DIAGNOSTICS

1. Magnetic probes

a) Principle of operation. Fluctuating RF magnetic fields inside the plasma can be measured with a magnetic probe, which is a small coil of wire, perhaps 2 mm in diameter, covered with glass or ceramic so as to protect it from direct exposure to charged particles.

When the coil is placed in a time-varying magnetic field **B**, an electric field is induced along the wire according to Faraday's Law:

$$\nabla \times \mathbf{E} = -d\mathbf{B}/dt\,. \tag{32}$$

Integrating this over the surface enclosed by the coil with the help of Stokes' theorem to convert the surface integral to a line integral, we obtain

$$\int \dot{\mathbf{B}} \cdot dS \equiv \dot{\Phi} = -\int (\nabla \times \mathbf{E}) \cdot dS = -\int \mathbf{E} \cdot d\ell \equiv -V_{ind}\,. \tag{33}$$

Here the line integral is along the wire in the coil and Φ is the magnetic flux through the coil, which is $\approx BA$, where A is the area of the coil. The induced voltage $V_{\rm ind}$ is measured by a high-impedance device like an oscilloscope. If there are N turns in the coil, the voltage will be N times higher; hence,

$$V_{ind} = -NA\dot{B} \tag{34}$$

The dot indicates the time-derivative and is the origin of the name "B-dot probe." The minus sign indicates that the induced electric field is in the opposite direction to that obtained when the right-hand rule is applied to **B**. For a sinusoidal signal, B-dot is proportional to ωB, so that the probe is more sensitive to higher frequencies. To obtain B from the measured $V_{\rm ind}$, one can use a simple integrator consisting of a resistor and a capacitor to ground to obtain

$$B = -\frac{1}{NA}\int V_{ind}\,dt\,. \tag{35}$$

One only has to be sure that the RC time constant of the integrator is much longer than the period of the signal.

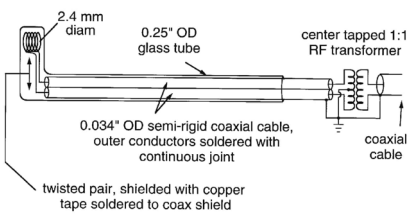


Fig. 30. A magnetic probe with a balun transformer.

b) Construction. The probe itself can be as simple as ten turns of thin wire wound on a core machined out of boron nitride. The coil can be placed inside a ceramic tube or a closed glass tube. Such a tube is necessarily larger than a Langmuir probe shaft and may disturb the plasma downstream from the source. If the axis of the coil is parallel to the tube, the component of **B** parallel to the shaft will be measured. If the coil axis is perpendicular to the shaft, one can change from $B_{\rm r}$ to B_θ measurement by rotating the shaft by 90°. Sometimes three coils are mounted in the same shaft to measure all three B components at the same time.

The difficult part is to take the signal out through the probe shaft without engendering too much RF pickup. One way is to use a very thin rigid coax, which is then connected to the scope with a 50-Ω cable. The coil in this case can be a single turn formed from the center conductor looped around and soldered to the conducting shield. If the shaft has to traverse a long path through the plasma, a better way is to use a multi-turn coil to increase the signal voltage, and then bring the two ends of the coils through the shaft with a twisted pair of wires. Outside the plasma, the wires are connected to a balun (balanced-to-unbalanced) 1-to-1 transformer so that the signal can be carried to the scope with an unbalanced line. Such a probe is shown in Fig. 30. The transformer can also have a turns ratio that amplifies the signal voltage. With magnetic probes there is always the danger of capacitive pickup through the insulators. One can check this by rotating the probe 180°. The magnetic signal should be the same in magnitude but shifted 180° in phase, while the capacitive signal would be the same in both orientations. Whether or not the probe and leads should be shielded with slotted conductors is a matter of experimentation; the shield can help or actually make the pickup worse.

2. Energy analyzers

Gridded energy analyzers are used to obtain better data for ion and electron energy distributions than can be obtained with Langmuir probes. However, these instruments are necessarily large—at least 1 cubic centimeter in volume—and will disturb the plasma downstream from them. A standard gridded analyzer has four grids: 1) a grounded or floating outer grid to isolate the analyzer from the plasma, 2) a grid with positive or negative potential to repel the unwanted species, 3) a solid collector with variable potential connected to the current measuring device, and 4) a suppressor grid in front of the collector to repel secondary electrons. In Fig. 31, *s* is the sheath edge. Grid G1, whether floating or grounded, will be negative with respect to the plasma, and therefore will repel all electrons except the most energetic ones. One cannot bias this grid positively, since it will then draw so much electron current that the plasma will be disturbed. It is sometime omitted in order to allow slower electrons to enter the analyzer. Grid G1 also serves to attenuate the flux of plasma into the analyzer so that the Debye length is not so short there that subsequent grid wires will be shielded out. In the space be-

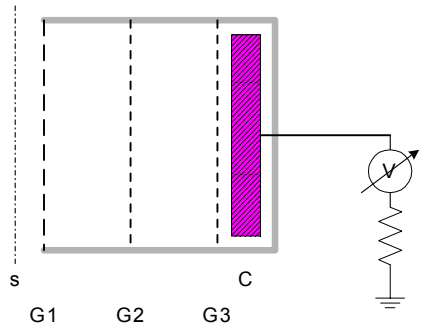


Fig. 31. A gridded energy analyzer.

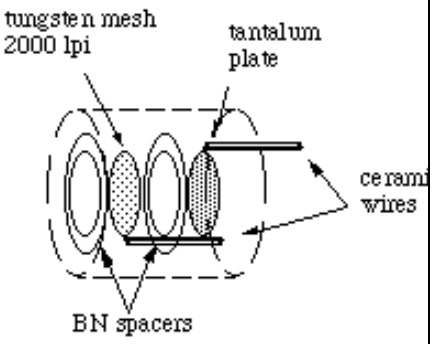


Fig. 32. Energy analyzer with only one grid and a collector

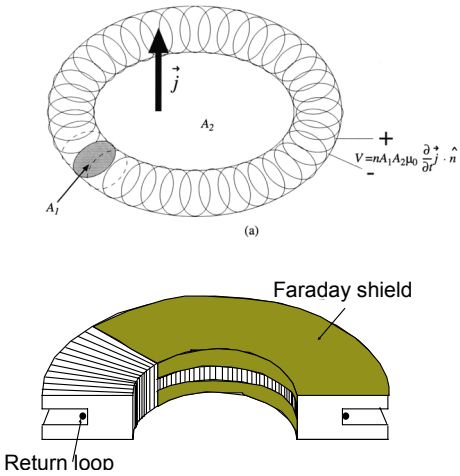


Fig. 33. Construction of an RF current probe.

hind Grid G1, there will be a distribution of ions which have been accelerated by the sheath but which still has the original relative energy distribution (unless it has been degraded by scattering off the grid wires). These are neutralized by electrons that have also come through G1. These electrons also have the original relative energy distribution, but they all have been decelerated by the sheath. Grid G2 is set positive to repel ions and negative to repel electrons. For example, to obtain $f_i(v)$, we would set G2 sufficiently negative (V_2) to repel all the electrons. The ions will then be further accelerated toward the collector. This collecting plate C, at V_c, would collect all the ion current if it were at the same potential as V_2. By biasing it more and more positive relative to V_2, only the most energetic ions would be collected. The curve of I vs. V_c would then give $f_i(v)$ when it is differentiated. When ions strike the collector, secondary electrons can be emitted, and these will be accelerated away from the collector by the field between C and G2, leading to a false enhancement of the apparent ion current. To prevent this, Grid G3 is fixed at a small negative potential (about −2V) relative to C) so that these electrons are turned back. Variations to this standard configuration are also possible.

In a plasma with RF fluctuations, energy analyzers would suffer from nonlinear averaging, just as Langmuir probes do. Because of their large size, and therefore stray capacitance, it would not be practical to drive the grids of an energy analyzer to follow changes in plasma potential at the RF frequency. However, one can design the circuitry to be fast enough to follow the RF and then record the oscillations in collected current as a function of time during each RF cycle. By selecting data from the same RF phase to perform the analysis, one can, in principle, obtain the true energy distribution. This technique cannot be used for Langmuir probes, because the currents there are so small that the required frequency response cannot be obtained. RF-sensitive energy analyzers have been made successfully by at least two groups; one such analyzer is shown in Fig. 32.

3. RF current probe

Current probes, sometimes called Rogowski coils, are coils of wire wound on a toroidal coil form shaped like a Life Saver. Figure 33 shows such a coil. Current passing through the hole induces a magnetic field in the azimuthal direction, and this, in turn, induces a voltage in the turns of wire. The current driven through the wire is

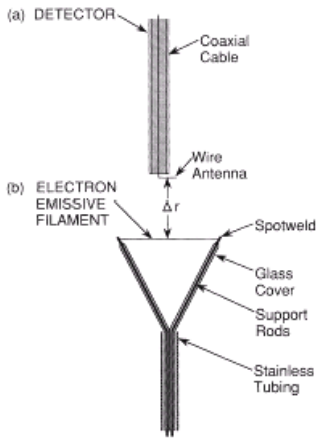


Fig. 34. Schematic of a POP [Shirakawa and Sugai, Jpn. J. Appl. Phys. **32**, 5129 (1993)].

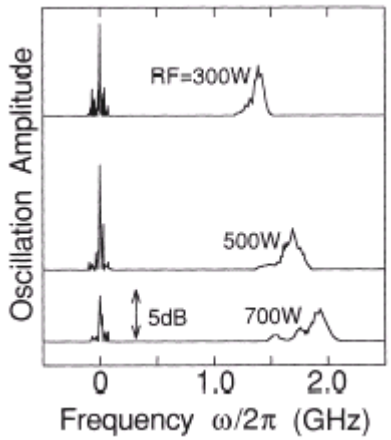


Fig. 35. Peak at ω_p (at the right) moves as P_{rf} is increased [*ibid.*].

then measured in an external circuit. The coil must take a return loop the long way around the torus to cancel the B-dot pickup that is induced by B-fields that thread the hole. Current probes are usually large and can be bought as attachments to an oscilloscope, but these are unsuitable for insertion into a plasma. The probe shown here is not only small (~1 cm diam) but is also made for RF frequencies. It is covered with a Faraday shield to reduce electrostatic pickup, and the windings are carefully calibrated so that the B-dot and E-dot signals are small compared with the J-dot signal. An example of a J-dot measurement was shown in Fig A6-17.

4. Plasma oscillation probe

When used in a plasma processing reactor, Langmuir probes tend to get covered with insulating coatings so that they can no longer properly measure dc current. A plasma oscillation probe avoids this by measuring only ac signals, which can pass capacitively through the coatings. A filament, like a hot probe (Fig. 34), is heated to emission and biased to ~100V negatively to send an electron beam into the plasma. Such a beam excites plasma waves near ω_p. These high-frequency oscillations are picked up by a probe and observed on a spectrum analyzer. If a peak in the response can be detected (Fig. 35), it will likely be near $\omega = \omega_p$, and this gives the plasma density. Various spurious effects, such as multiple peaks or surface waves, can cause the resonant ω to differ from ω_p, but when the signal is clear, a good estimate of n can be obtained.